

AIC372 — Knowledge Representation and Reasoning

Lecture 2: Knowledge-Based Systems, Logic, Reasoning and AI

Program: BS Computer Science

Course Code: AIC372

Unit: 1

Lecture: 2

Duration: 75 Minutes

Reference: Levesque, Chapter 1

Syllabus Topics

1. Knowledge-Based Systems
2. Why Knowledge Representation?
3. Why Reasoning?
4. The Role of Logic
5. Relationship between Knowledge and Artificial Intelligence

The lecture continues the conceptual foundation established in Lecture 1 and prepares students for the formal knowledge representation and reasoning techniques studied later in the course.

1. CLO and Bloom's Taxonomy Alignment

The syllabus identifies **CLO-1** as:

Explain types of knowledge and their relation to intelligence.

CLO-1 belongs to **Unit 1**, is classified at the **Understanding** level of Bloom's Taxonomy, and is associated with **SO1**.

Lecture 2 Alignment

Component	Alignment
CLO	CLO-1
CLO Statement	Explain types of knowledge and their relation to intelligence
CLO Bloom Level	Understand
Student Outcome	SO1

Component	Alignment
Primary Lecture Bloom Level	Understand
Secondary Bloom Levels	Apply, Analyze
Knowledge Dimension	Conceptual
Main Assessment	Explanation, comparison, system analysis, simple logical inference

2. Lecture Learning Outcomes

By the end of this lecture, students will be able to:

SLO 1 — Understand

Define a Knowledge-Based System and identify its major components.

SLO 2 — Understand

Explain why explicit knowledge representation is important in Artificial Intelligence.

SLO 3 — Understand

Explain why reasoning is necessary for an intelligent system.

SLO 4 — Understand

Explain the role of logic in representing and reasoning about knowledge.

SLO 5 — Apply

Apply simple logical rules to derive conclusions from represented knowledge.

SLO 6 — Analyze

Analyze the relationship between knowledge, reasoning, logic, and intelligent behavior in an AI system.

3. Bloom's Taxonomy Alignment

The main cognitive level for this lecture is **Understand**, because the official CLO is at the Understanding level.

However, students will progressively move toward:

Understand → Apply → Analyze

Bloom Level	Lecture 2 Activity
Remember	Recall definitions
Understand	Explain KBS, logic, reasoning
Apply	Apply simple rules
Analyze	Analyze an AI system's architecture

4. 75-Minute Teaching Plan

Time	Topic	Bloom Level
0–5 min	Review of Lecture 1	Remember / Understand
5–18 min	Knowledge-Based Systems	Understand
18–30 min	Components of KBS	Understand
30–40 min	Why Knowledge Representation?	Understand / Analyze
40–50 min	Why Reasoning?	Understand / Apply
50–62 min	Role of Logic	Understand / Apply
62–68 min	Relationship between Knowledge and AI	Analyze
68–73 min	Classroom Activity	Apply / Analyze
73–75 min	Assessment and Summary	Understand

5. Opening Review

Time: 0–5 minutes

Start by asking students:

What did we learn in Lecture 1?

Expected answers:

- Knowledge
- Representation

- Reasoning
- Knowledge-based systems
- Intelligent behavior

Write:

```text id="2d8q4s" Knowledge ↓ Representation ↓ Reasoning ↓ Conclusion ↓ Intelligent Behavior

Then ask:

> If knowledge and reasoning are important for AI, how can we build an actual AI system around them?

This question leads directly to today's topic:

> **Knowledge-Based Systems**

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# 6. What Is a Knowledge-Based System?

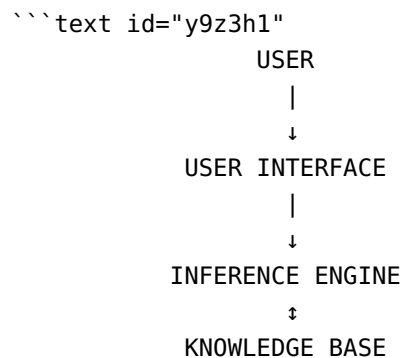
### Time: 5–18 minutes

A **Knowledge-Based System (KBS)** is an AI system that explicitly represents knowledge about a domain and uses reasoning mechanisms to derive conclusions or support problem solving.

In simple terms:

> A Knowledge-Based System stores knowledge and uses reasoning to make that knowledge useful.

A simplified architecture is:



The two most important components are:

1. **Knowledge Base**
  2. **Inference Engine**
- 

## 7. Knowledge Base

The **Knowledge Base (KB)** contains the knowledge of the system.

It may contain:

- Facts
- Rules
- Concepts
- Relationships
- Constraints
- Domain knowledge

### Example

Suppose we build a university advising system.

The knowledge base may contain:

```
```text id="h7x4yz" Student(Ali)
```

```
Studies(Ali, ComputerScience)
```

```
Completed(Ali, ProgrammingFundamentals)
```

```
Requires(DataStructures, ProgrammingFundamentals)
```

It may also contain rules.

For example:

```
```text id="j7r6uk"
If a student has completed
all prerequisites of a course,
then the student can register
for that course.
```

The knowledge base represents **what the system knows**.

---

## 8. Inference Engine

The **inference engine** is responsible for reasoning with the knowledge base.

It determines:

What conclusions can be derived from the available knowledge?

Suppose the KB contains:

```
```text id="f1k6j4" Student(Ali)
```

$\text{Student}(x) \rightarrow \text{HasID}(x)$

The inference engine can derive:

```
```text id="v5v9p2"
HasID(Ali)
```

Thus:

```
```text id="f5e6fk" Knowledge Base ↓ Inference Engine ↓ New Conclusion
```

```
---
```

9. User Interface

The user interface allows people to communicate with the knowledge-based system.

For example:

```
### User
```

```
> Can Ali register for Data Structures?
```

```
### System
```

```
> Yes. Ali has completed the required prerequisite.
```

The user does not need to know the internal representation.

The system handles the reasoning internally.

10. Complete KBS Example

Consider a simple medical advisory system.

Knowledge Base

Facts:

```
```text id="y8x7n6"
```

```
HasSymptom(Ali, Fever)
```

```
HasSymptom(Ali, Cough)
```

Rule:

```
```text id="9c6u8m" Fever(x) ∧ Cough(x) → PossibleRespiratoryInfection(x)
```

User

> What condition might Ali have?

Inference Engine

It examines:

```
```text id="zj73nb"
```

```
Fever(Ali)
```

```
Cough(Ali)
```

It finds a matching rule:

```
```text id="6z7nqf" Fever(x) ∧ Cough(x) → PossibleRespiratoryInfection(x)
```

It derives:

```
```text id="w4u9a7"
```

```
PossibleRespiratoryInfection(Ali)
```

### Important teaching point

The system did not necessarily store the conclusion directly.

It **derived** it.

That is the power of reasoning over represented knowledge.

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## 11. Why Knowledge Representation?

**Time: 30–40 minutes**

Now ask:

Why do we need a special representation of knowledge?

Humans can understand natural language relatively easily.

Computers require representations that can be processed systematically.

For example:

"All registered students must have a university ID."

We can represent the rule conceptually as:

```
```text id="9x4r3a" Registered(x) → HasID(x)
```

And:

```
```text id="4z5t6w"
Registered(Ali)
```

allows us to derive:

```
```text id="p2h7y3" HasID(Ali)
```

```
---
```

```
# 12. KR Provides Structure
```

```
Suppose we have this statement:
```

```
> Ali studies Computer Science at COMSATS.
```

A structured representation could distinguish:

```
```text id="c8b4y7"
```

```
Student(Ali)
```

```
Studies(Ali, ComputerScience)
```

```
EnrolledAt(Ali, University)
```

The advantage is that the relationships are explicit.

A reasoning mechanism can operate on them.

---

## 13. KR Allows Generalization

Consider:

Ali is a student.

and:

Sara is a student.

Instead of storing unrelated information, we can represent a general concept:

```
```text id="w2m8r6" Student(x)
```

and individual facts:

```
```text id="x4j8p9"
```

```
Student(Ali)
```

```
Student(Sara)
```

A general rule can then apply to both.

For example:

```
```text id="m8r2s4" Student(x) → HasID(x)
```

The system can infer:

```
```text id="z8a2k7"  
HasID(Ali)
HasID(Sara)
```

This is one reason structured representation is powerful.

---

## 14. KR Supports Reasoning

Without a structured representation, reasoning becomes difficult.

Suppose the system knows:

```
```text id="4q8k3m" Bird(Tweety)
```

$\text{Bird}(x) \rightarrow \text{Animal}(x)$

The system can derive:

```
```text id="v7m5z1"  
Animal(Tweety)
```

The representation gives the reasoning mechanism something precise to operate on.

---

## 15. Why Reasoning?

**Time: 40–50 minutes**

Now ask:

If we already have a knowledge base, why do we need reasoning?

Because storing knowledge is not enough.

An intelligent system often needs to derive information that is **not explicitly stored**.

---

## 16. Explicit vs Derived Knowledge

Suppose the knowledge base contains:

```
```text id="h8m1z9" Student(Ali)
```

$\text{Student}(x) \rightarrow \text{HasID}(x)$

The KB explicitly contains:

> Ali is a student.

It does not necessarily explicitly contain:

> Ali has an ID.

The inference process derives:

```
```text id="m7r4x2"
HasID(Ali)
```

Therefore:

**Reasoning allows an AI system to obtain new conclusions from existing knowledge.**

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## 17. Reasoning Reduces the Need to Store Every Fact

Imagine a university has:

- 10,000 students.

Suppose every student has an ID.

We could explicitly store:

```
HasID(Ali)
HasID(Ahmed)
HasID(Sara)
...
```

But if the general rule is:

```
```text id="u9s7e2" Student(x) → HasID(x)
```

then the system can derive the appropriate conclusion whenever needed.

This illustrates the usefulness of **general rules**.

18. Reasoning for Problem Solving

Consider a troubleshooting system.

Knowledge:

```
```text id="w6k3r8"
BatteryLow(Car1)
```

Rule:

```
```text id="m4x9q2" BatteryLow(x) → EngineMayNotStart(x)
```

Another rule:

```
```text id="b7n3c5"
EngineMayNotStart(x)
→ CheckBattery(x)
```

Reasoning produces:

```
```text id="c4p8v2" BatteryLow(Car1) ↓ EngineMayNotStart(Car1) ↓ CheckBattery(Car1)
```

The system has moved from a fact to an actionable recommendation.

19. The Role of Logic

Time: 50–62 minutes

Now introduce the most important conceptual tool in today's lecture.

```
> **Logic provides formal languages and principles for representing knowledge
and reasoning about what follows from that knowledge.**
```

In simple words:

```
> **Logic helps us represent statements precisely and determine valid
conclusions.**
```

20. Why Natural Language Is Difficult for Formal Reasoning

Consider:

```
> "Ali is a good student."
```

What does "good" mean?

Does it mean:

- High marks?
- Regular attendance?
- Good behavior?
- Excellent programming skills?

Natural language can be ambiguous.

Formal logic aims to reduce such ambiguity by using precise structures.

For example:

```
```text id="b2k5x8"
Student(Ali)
```

is more precise than saying:

Ali is a student.

within a formal reasoning system.

---

## 21. Logic as a Representation Language

Logic allows us to express:

## Facts

```
```text id="k6q1m8" Student(Ali)
```

```
### Relationships
```

```
```text id="q8w4j2"  
Studies(Ali, ComputerScience)
```

## Rules

```
```text id="t5r9n3" Student(x) → HasID(x)
```

```
### Conclusions
```

```
```text id="e3v7c1"  
HasID(Ali)
```

These structures can be processed by reasoning mechanisms.

---

## 22. A Simple Logical Inference

Write on the board:

```
```text id="7g3w9q" Rule: Human(x) → Mortal(x)
```

Fact: Human(Socrates)

Conclusion: Mortal(Socrates)

Explain:

1. The rule applies to every human.
2. Socrates is a human.
3. Therefore, Socrates is mortal.

This illustrates how logic supports formal reasoning.

23. Logic and Correct Reasoning

Logic provides principles for determining whether a conclusion follows from premises.

Consider:

```
```text id="x5c8m2"
All birds have wings.
Tweety is a bird.
Therefore, Tweety has wings.
```

The conclusion follows from the premises.

But consider:

```
```text id="r8v2n6" All birds have wings. Tweety has wings. Therefore, Tweety is a bird.
```

This conclusion does **not necessarily follow**.

Why?

Because many things other than birds could have wings.

This is an important lesson:

> **Reasoning must respect the structure of the knowledge and the rules being used.**

24. Logic Helps Separate Valid and Invalid Inferences

Consider:

Argument A

```
```text
All humans are mortal.
Ali is a human.
Therefore Ali is mortal.
```

Valid.

### Argument B

All humans are mortal.  
Ali is mortal.  
Therefore Ali is human.

Not necessarily valid.

The second argument reverses the implication incorrectly.

This is why formal reasoning matters.

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## 25. Relationship Between Knowledge and AI

**Time: 62–68 minutes**

Now connect today's concepts to Artificial Intelligence.

AI aims to create systems that exhibit intelligent behavior.

Intelligent behavior may include:

- Problem solving
- Decision making
- Planning
- Question answering
- Diagnosis
- Understanding situations
- Selecting actions

Knowledge plays an important role in these activities.

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## 26. Knowledge as a Foundation for Intelligent Behavior

Consider an autonomous vehicle.

The system needs knowledge about:

- Roads
- Traffic rules

- Vehicles
- Pedestrians
- Traffic signals
- Safe distances
- Possible actions

It also needs reasoning.

For example:

```
```text id="j6v2r9" PedestrianInPath ↓ PotentialCollision ↓ ReduceSpeed
```

Thus:

```
```text id="k3x7p8"
Knowledge
 +
Reasoning
 ↓
Decision
 ↓
Action
```

---

## 27. Knowledge and AI Applications

### Medical AI

Knowledge:

- Symptoms
- Diseases
- Treatments
- Relationships

Reasoning:

Which diagnosis is consistent with the observed symptoms?

---

### Intelligent Tutoring

Knowledge:

- Student performance

- Course concepts
- Prerequisites
- Learning rules

Reasoning:

What should the student study next?

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## **Robotics**

Knowledge:

- Objects
- Locations
- Actions
- Constraints

Reasoning:

Which action should the robot perform?

---

## **Autonomous Vehicles**

Knowledge:

- Traffic rules
- Road conditions
- Obstacles
- Vehicle states

Reasoning:

What action should be taken next?

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# **28. Knowledge-Based AI vs Pure Data Processing**

This distinction is important.

A conventional data-processing system may answer:

How many students obtained more than 80 marks?

A knowledge-based system may answer:

Which students are eligible for a particular course based on their completed prerequisites?

The second problem requires understanding relationships and applying rules.

Therefore, knowledge representation and reasoning are particularly useful when AI systems need to operate with:

- Rules
  - Relationships
  - Concepts
  - Constraints
  - Conclusions
- 

## 29. Classroom Activity

**Time: 68–73 minutes**

### **Activity: Design a Mini Knowledge-Based System**

Divide the class into groups.

Give each group this scenario:

#### **University Course Registration System**

Ask students to identify:

##### **A. Facts**

Example:

```
Student(Ali)
```

```
Completed(Ali, Programming)
```

##### **B. Rule**

```
Completed(x, Programming)
→ EligibleFor(x, DataStructures)
```

### C. Conclusion

```
EligibleFor(Ali, DataStructures)
```

### D. Identify the components

Students should identify:

- Knowledge Base
- Inference Engine
- User
- Question
- Conclusion

---

## 30. Discussion Questions for the Activity

Ask:

### Question 1

Which information is knowledge?

### Question 2

Which part is the rule?

### Question 3

Which part is the reasoning?

### Question 4

Which conclusion was not explicitly given?

### Question 5

How does this system demonstrate intelligent behavior?

---

## 31. Bloom's Taxonomy Assessment

Use the following questions to assess different cognitive levels.

## Remember

What is a Knowledge-Based System?

Expected:

A system that represents domain knowledge and uses reasoning mechanisms to derive conclusions or solve problems.

---

## Understand

Explain the difference between a knowledge base and an inference engine.

Expected:

The knowledge base stores represented knowledge, while the inference engine reasons over that knowledge.

---

## Apply

Given:

````text id="q5j2w7" Student(Ali)`

$\text{Student}(x) \rightarrow \text{HasID}(x)$

Ask:

> What can be inferred?

Answer:

````text id="x9p4k6"  
HasID(Ali)`

---

## Analyze

Ask:

Analyze how a university advising system uses knowledge, representation, logic, and reasoning to recommend a course.

Students should identify:

- Knowledge
  - Representation
  - Rules
  - Inference
  - Conclusion
  - Decision
- 

## 32. Common Misconceptions

### Misconception 1

A knowledge base and database are exactly the same.

#### Correction

A database primarily stores structured data. A knowledge base may contain richer conceptual knowledge, rules, relationships, and other information intended for reasoning.

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### Misconception 2

The inference engine stores all the knowledge.

#### Correction

The knowledge base stores the represented domain knowledge. The inference engine operates on it.

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### Misconception 3

Logic automatically makes an AI intelligent.

#### Correction

Logic is a tool for representation and reasoning. Intelligent behavior also depends on the knowledge available, the reasoning mechanism, goals, perception, actions, and other components.

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## Misconception 4

More knowledge always means better AI.

### Correction

Knowledge must be relevant, appropriately represented, and usable for reasoning.

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## Misconception 5

If a conclusion sounds reasonable, it must logically follow.

### Correction

A conclusion can sound plausible without being logically entailed by the premises.

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# 33. CLO-Based Formative Assessment

## CLO-1

**Explain types of knowledge and their relation to intelligence.**

### Task

Consider this scenario:

A university AI advisor knows the courses completed by a student and the prerequisites of available courses. It uses rules to determine which courses the student can register for.

Ask students to explain:

1. What knowledge does the system possess?
2. How is that knowledge represented?
3. What role does reasoning play?
4. Why is logic useful?
5. How does the system demonstrate intelligent behavior?

### Expected learning evidence

Students should be able to explain the chain:

DOMAIN KNOWLEDGE ↓ KNOWLEDGE REPRESENTATION ↓ LOGICAL RULES ↓  
REASONING ↓ CONCLUSION ↓ INTELLIGENT DECISION

---

## # 34. Homework

### ## Task 1 – Conceptual

Define and explain:

1. Knowledge-Based System
2. Knowledge Base
3. Inference Engine
4. Knowledge Representation
5. Reasoning
6. Logic

---

### ## Task 2 – Application

Consider a **smart home system**.

Identify:

1. Three facts.
2. Two rules.
3. One possible conclusion.
4. The role of the inference engine.

Example:

```text

MotionDetected(Room1)

Dark(Room1)

MotionDetected(x) $\wedge$ Dark(x)

$\rightarrow$ TurnOnLight(x)

Conclusion:

TurnOnLight(Room1)

Task 3 — Analysis

Choose one AI application and explain:

How does knowledge contribute to intelligent behavior in that application?

Possible applications:

- Robotics
 - Medical diagnosis
 - Autonomous driving
 - University advising
 - Smart home
 - Intelligent tutoring
-

35. End-of-Lecture Exit Ticket

Ask students to answer these three questions.

Q1

What are the two core components of a basic Knowledge-Based System?

Answer:

- Knowledge Base
 - Inference Engine
-

Q2

Why do we need reasoning?

Answer:

To derive new conclusions from existing knowledge and support problem solving or decision making.

Q3

What is the role of logic?

Answer:

Logic provides formal methods for representing knowledge and determining what conclusions follow from that knowledge.

36. Key Takeaways

Students should remember:

1.

A **Knowledge-Based System** explicitly represents knowledge and reasons with it.

2.

The **Knowledge Base** contains represented domain knowledge.

3.

The **Inference Engine** performs reasoning over the knowledge.

4.

Knowledge representation provides structure that allows reasoning.

5.

Reasoning derives conclusions that may not be explicitly stored.

6.

Logic provides a formal foundation for representing knowledge and determining valid conclusions.

7.

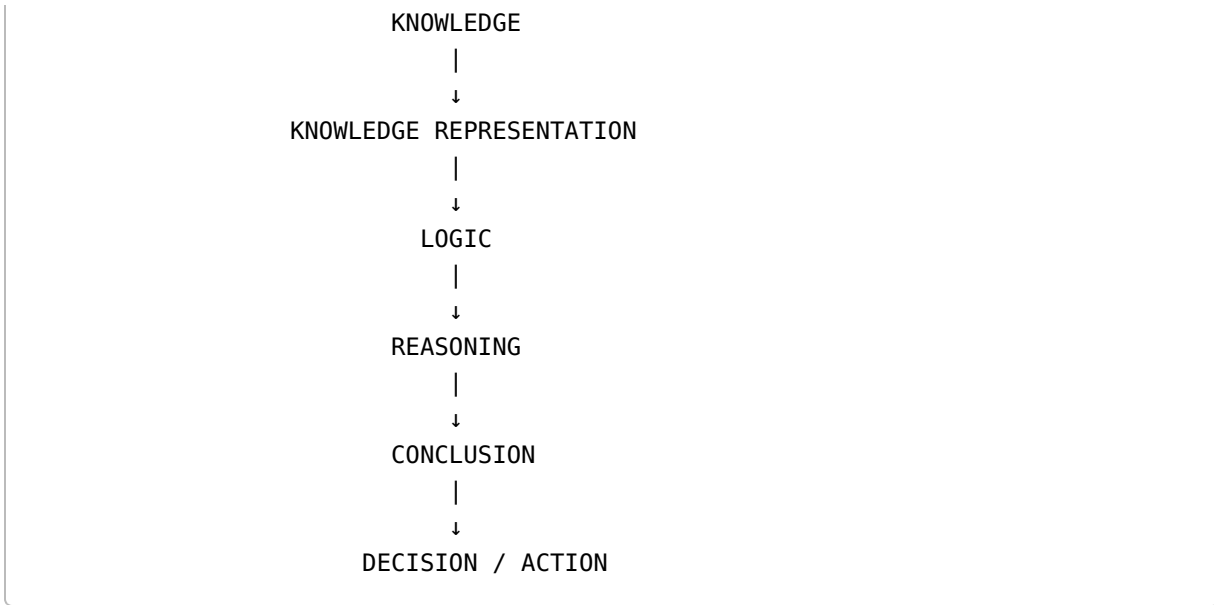
Knowledge and reasoning contribute to intelligent behavior.

37. Final Conceptual Model

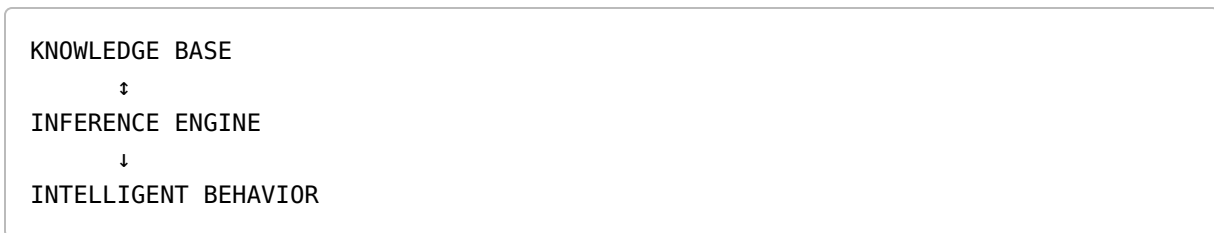
Write the following on the board:

ARTIFICIAL INTELLIGENCE





Then write:



38. Final CLO-Bloom-Assessment Matrix

| SLO | CLO | Bloom Level | Assessment |
|---|-------|----------------------|---------------------|
| Define KBS | CLO-1 | Understand | Short question |
| Explain components of KBS | CLO-1 | Understand | Conceptual question |
| Explain why KR is needed | CLO-1 | Understand | Written response |
| Explain why reasoning is needed | CLO-1 | Understand | Discussion |
| Apply simple logical rule | CLO-1 | Apply | Worked problem |
| Analyze AI application | CLO-1 | Analyze | Scenario-based task |
| Explain relationship between knowledge and intelligence | CLO-1 | Understand / Analyze | CLO assessment |

39. Instructor Closing Statement

"In Lecture 1, we introduced knowledge, representation, and reasoning. Today we connected these concepts to actual Knowledge-Based Systems. We learned that a knowledge base stores represented knowledge, while an inference engine reasons over that knowledge. We also saw why logic is important: it gives us a formal way to represent information and determine what conclusions follow. In the next lecture, we will move from these foundations toward the practical process of **Knowledge Engineering**, including vocabulary and different kinds of facts."